

AIM:

To create a C program using multiple source files, header file, macros, conditional compilation, and extern variables to calculate and display the area of a square or a circle based on the value of a macro.

PROCEDURE

1. Open Notepad and create a header file named main.h.
2. Define the macro STAGE and conditional compilation to define the AREA macro for square or circle.
3. Create a source file main.c and include the header file.
4. Declare global variables using conditional compilation based on STAGE.
5. Create another source file display.c.
6. Use the extern keyword to access global variables declared in main.c.
7. Print the calculated area using the AREA macro.
8. Compile the program using GCC.

PROGRAM:

- **main.h**

```
/**
 * Header file containing macro definitions and AREA macro
 * This file defines the STAGE macro and conditional compilation
 * and defines the AREA macro for square or circle.
 * @file main.h
 * Date: 09-01-2026
 */
#ifndef MAIN_H
#define MAIN_H
#include <stdio.h>
#define STAGE 1
#if STAGE == 1
    #define AREA (side * side)
#elif STAGE == 2
    #define AREA (3.14 * radius * radius)
#endif
void display(void);
#endif
```

- **main.c**

```
/**
 * Main source file containing global variables and main function
 * This file defines global variables based on STAGE
 * and calls the display function.
 * @file main.c
 * Date: 09-01-2026
 */

#include "main.h"
#if STAGE == 1
int side = 5;      /** Side of the square */
#elif STAGE == 2
float radius = 3.0; /** Radius of the circle */
#endif
int main(void)
{
    display();
    return 0;
}
```

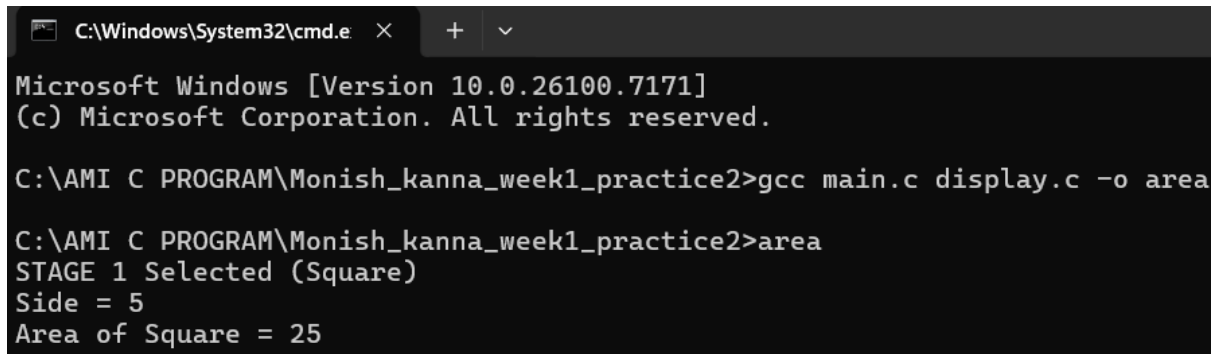
- **display.c**

```
/**
 * Source file to display area using AREA macro
 * This file accesses global variables using extern
 * and prints the calculated area.
 * @file display.c
 * Date 9-1-2026
 */

#include "main.h"
#if STAGE == 1
extern int side;
#elif STAGE == 2
extern float radius;
#endif
void display(void)
{
    #if STAGE == 1
        printf("STAGE 1 Selected (Square)\n");
        printf("Side = %d\n", side);
        printf("Area of Square = %d\n", AREA);
    #endif
}
```

```
#elif STAGE == 2
    printf("STAGE 2 Selected (Circle)\n");
    printf("Radius = %.2f\n", radius);
    printf("Area of Circle = %.2f\n", AREA);
#endif
}
```

OUTPUT:



```
C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26100.7171]
(c) Microsoft Corporation. All rights reserved.

C:\AMI C PROGRAM\Monish_kanna_week1_practice2>gcc main.c display.c -o area

C:\AMI C PROGRAM\Monish_kanna_week1_practice2>area
STAGE 1 Selected (Square)
Side = 5
Area of Square = 25
```

RESULT:

Thus, the C program using header files, macros, conditional compilation, and multiple source files was successfully compiled and executed.